Group meeting [January 28th 2026]

Parties present:

Ryan
Howard
Soroush
Alex
Claire
David (on Zoom)

Agenda made by:

Alex

Minutes taken by:

Ryan

Meeting minutes:

Do we need to trim evenly?

- No, reverse and forward reads do not need to be trimmed evenly. Reverse reads are shorter. We want 10-20 base pairs of overlap for the forward and reverse reads.
- Use the 515 and 806 primers. The max length of read for Illumina is 250.

How should we break up our data into groups?

- HA or/and NA. Look into which one has more influence on the microbiome if we do need to choose between the two.
- Current subtypes with greater than 5 samples:
- H5N5, H5N2, H5N1, H4N6 H3N8, H2N3.

Start Analysis by HA & NA

- If no clusters, we can try combining subtypes of similar HAs or NAs.

Outputs we need: table, taxonomy, phylogeny, and metadata file from Qiime to input into R for beta diversity analysis.

Figure out the rarefaction depth

Demux.qzv for 1C.

Photos form white board

How influenza affects the gut microbiome

IN R

[neg cont]

1500 bases

H5N5
H5N2
H5N1
H4N6
H3N8
H7N3

ST

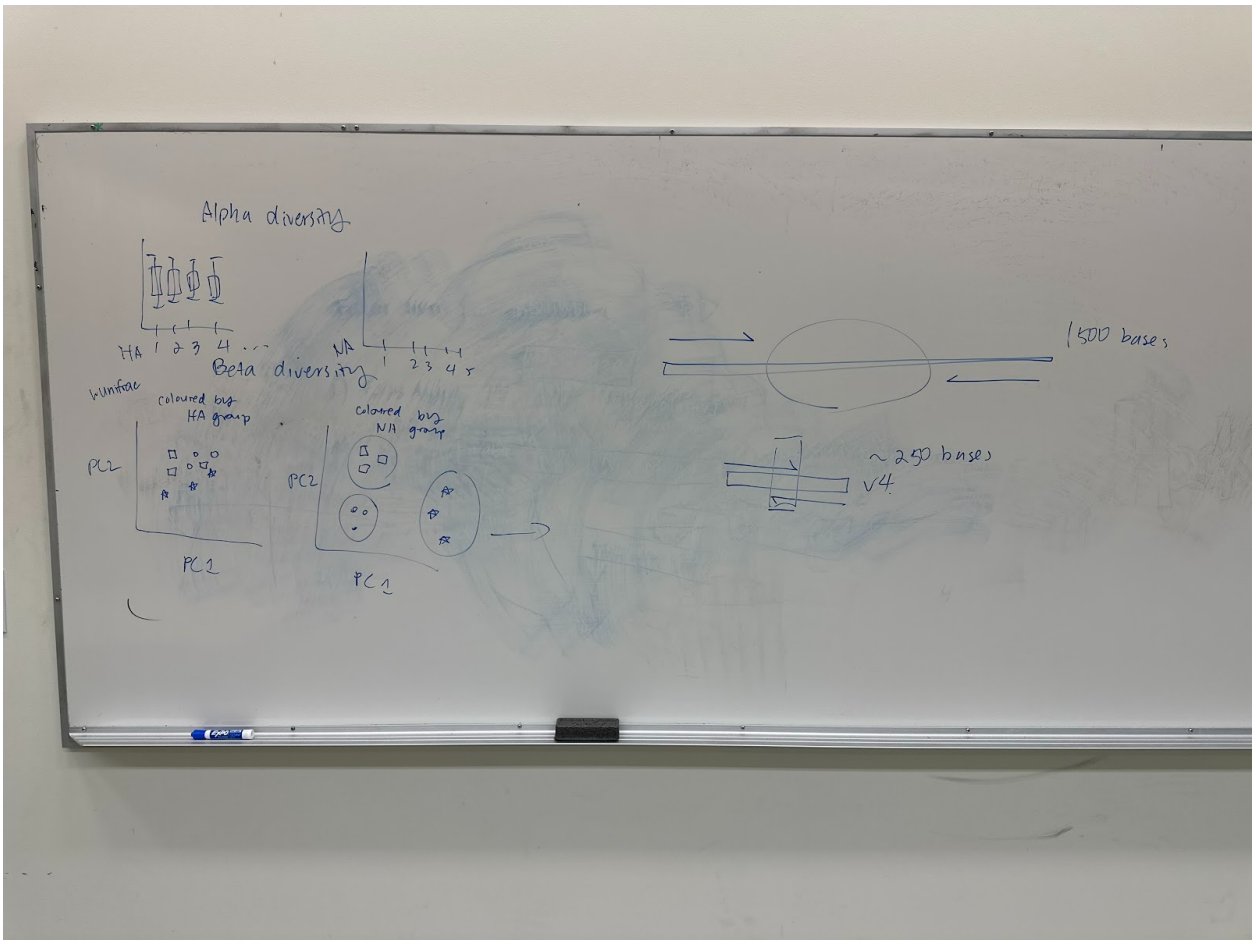
by subtype

Start analisies by HA & NA
↳ if no clusters seen go general.

Outputs

1) Table.qza
2) taxon.qza
3) phylo tree.qza
repseq.qza → maybe.
4) metadata.

Do β & α div in R.



Action Items:

-Generate a rarefaction curve
-Figure out the rarefaction depth
-Figure out the taxonomy file
-Adjust the trimming?
-Do some reading on how other papers have presented and analyzed data?
-Look at how the different influenza subtypes influence the gut microbiome
- Sign up for GitHub and send it to Alex.
-Finish the group project
-Define the functions of HA and NA for the assignment (Ryan + anyone who took MICB 306)